

Vendor Performance Analysis

An evidence-led case study of sales concentration, brand economics, purchase behaviour, and inventory signals.

Executive summary

SALES

\$450.44M

GROSS PROFIT

\$129.53M

GROSS MARGIN

28.8%

VENDORS

127

The pipeline consolidates 10,648 vendor–brand purchase records into a decision-ready view. Revenue is concentrated: the ten largest vendors account for 65.2% of recorded sales. At the record level, 2,106 observations carry negative gross profit and 178 have no sales dollars.

Portfolio framing: this is an analytical decision-support product, not an accounting close. Freight is attached at vendor level in the source pipeline and is intentionally excluded from product-level gross-profit aggregation.

Business question

VERIFIED RESULT

Which suppliers and products merit attention?

INTERPRETATION

The project combines transactions, product pricing and vendor freight into a unified analytical table.

BUSINESS IMPLICATION

Prioritize commercial action by revenue, margin and movement rather than viewing transactions in isolation.

LIMITATION

The period boundary and operational service levels are not specified in the source data.

From raw files to decisions

VERIFIED RESULT

Six operational extracts become one 18-column summary table.

INTERPRETATION

CSV extracts are ingested into SQLite; SQL aggregates purchases, sales and freight; pandas derives margin, turnover and sales-to-purchase ratios.

BUSINESS IMPLICATION

A local, repeatable workflow keeps analysis portable and makes each calculation inspectable.

LIMITATION

A vendor-level freight total is joined to every vendor–brand record and must not be summed across brands.

Data profile & quality

VERIFIED RESULT

10,648 summary rows, 127 vendors and 10,648 brands were analysed.

INTERPRETATION

The stored main notebook confirms the 18-column summary shape. Zero sales rows are retained by the left join, preserving possible slow-moving inventory signals.

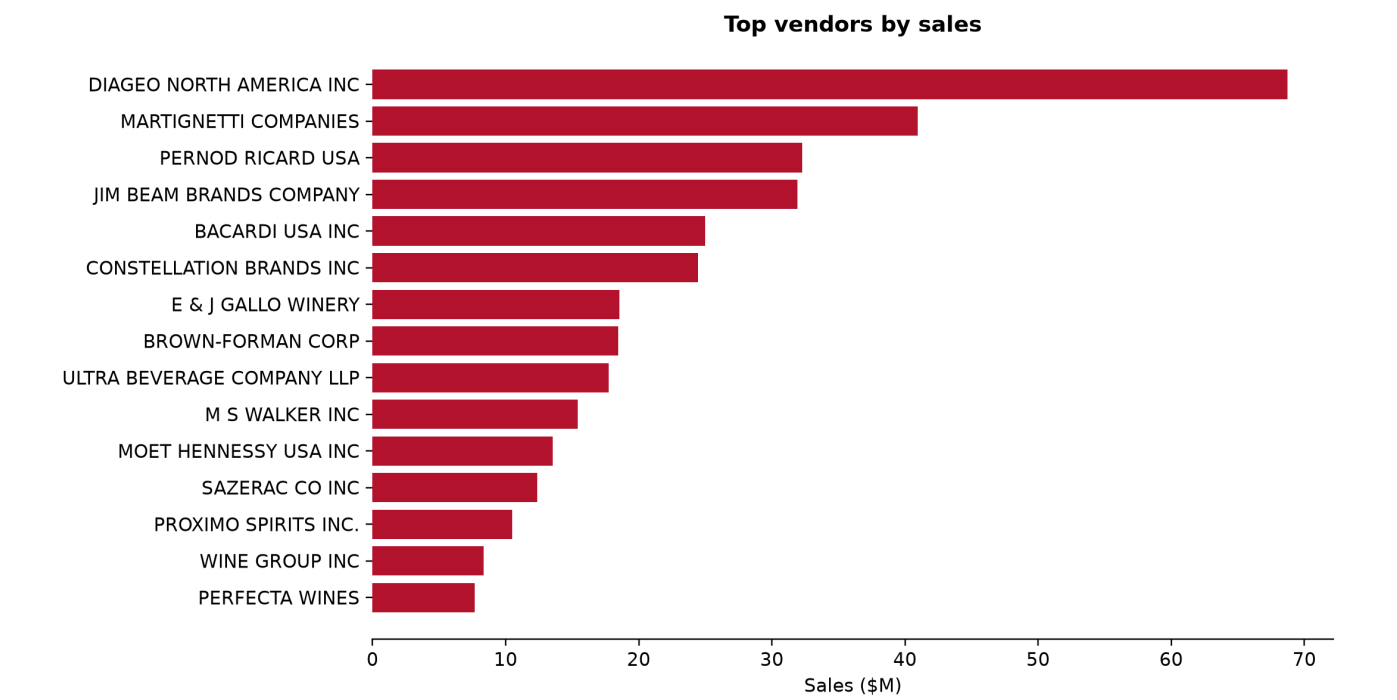
BUSINESS IMPLICATION

Preserve no-sale items for replenishment review; do not silently filter them out of operational monitoring.

LIMITATION

Zero filling enables ratios but cannot distinguish a true zero from every upstream reason for missing sales.

EDA: vendor sales landscape



Additional chart generated directly from vendor_sales_summary; bars show aggregate sales by vendor.

VERIFIED RESULT
DIAGEO NORTH AMERICA INC is the largest recorded vendor by sales at \$68.74M.

INTERPRETATION The sales distribution is visibly long-tailed, so averages alone mask where commercial exposure sits.	BUSINESS IMPLICATION Use vendor segmentation to focus relationship management and forecasting effort.	LIMITATION Sales identifies scale, not net profitability after fully allocated operating costs.
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KPI scorecard

VERIFIED RESULT

Recorded sales are \$450.44M against \$320.91M of purchase dollars, yielding a 28.8% gross margin.

INTERPRETATION

The KPI is calculated as total sales less recorded purchase dollars divided by sales.

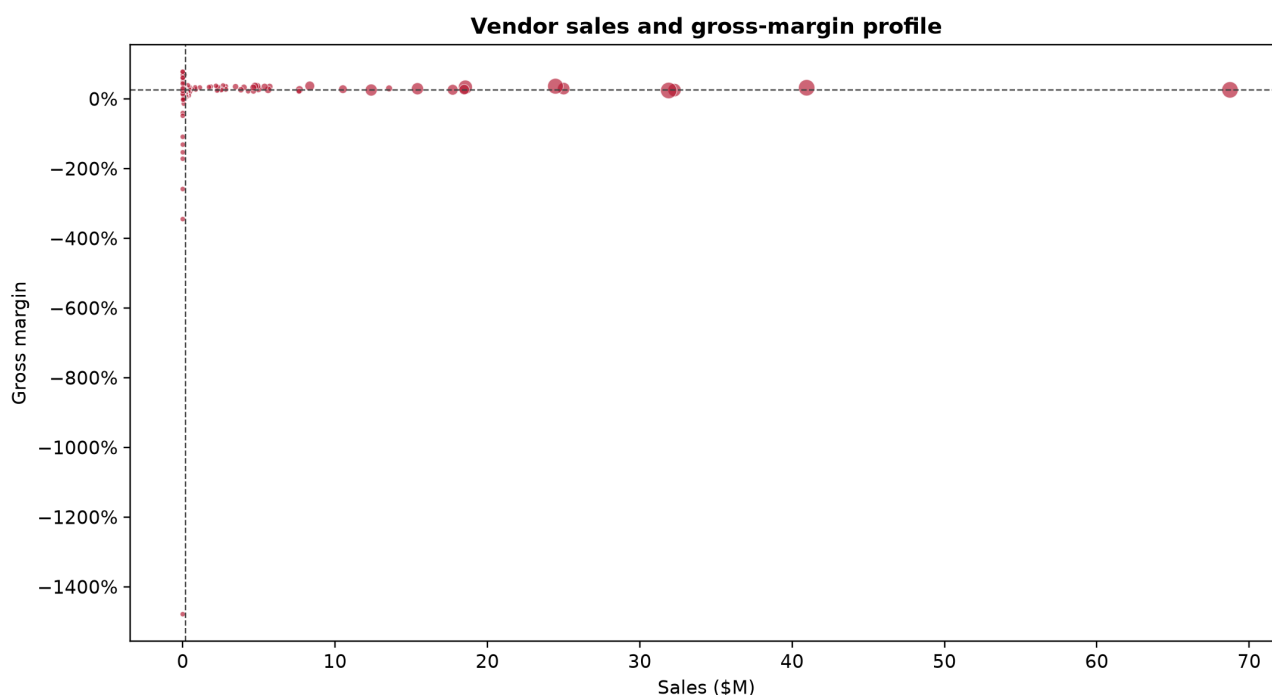
BUSINESS IMPLICATION

Use this as a directional commercial margin screen and pair it with cost-to-serve before pricing decisions.

LIMITATION

Freight, tax treatment, overhead and timing effects are not allocated at product level.

Vendor economics



Bubble size reflects purchase quantity; dashed lines mark median vendor sales and aggregate margin.

VERIFIED RESULT

The vendor portfolio spans distinct sales and gross-margin positions.

INTERPRETATION

High-scale, low-margin vendors and smaller, higher-margin vendors require different commercial interventions.

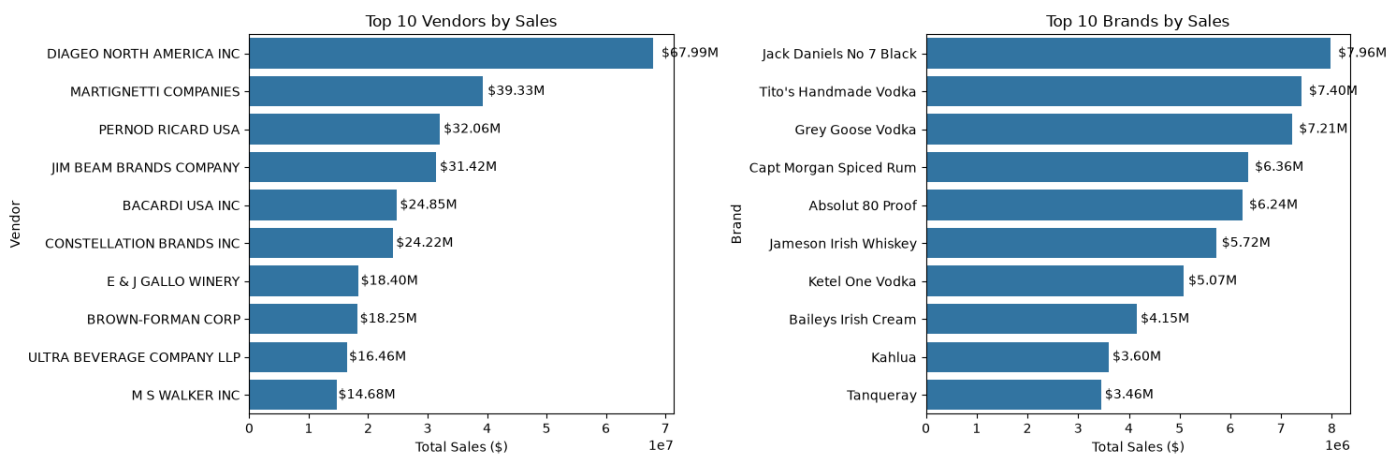
BUSINESS IMPLICATION

Set account plans by quadrant: protect profitable scale; investigate large low-margin exposure.

LIMITATION

The visual is descriptive; it does not establish that vendor choice causes margin outcomes.

Existing project visual



Reused project chart: top ten vendors and brands ranked by sales.

VERIFIED RESULT

The leading brand is Jack Daniels No 7 Black at \$5.10M in recorded sales.

INTERPRETATION

Product concentration accompanies vendor concentration, increasing exposure to a small number of commercial levers.

BUSINESS IMPLICATION

Protect availability and pricing discipline for sales-leading brands while checking their margin contribution.

LIMITATION

The original figure ranks sales only; it does not include returns, stock-outs or customer demand elasticity.

Brand portfolio

VERIFIED RESULT

The summary contains 10,648 brands; the top brand contributes \$5.10M in sales.

INTERPRETATION

Brand totals are aggregated from the same vendor–brand summary grain used throughout this study.

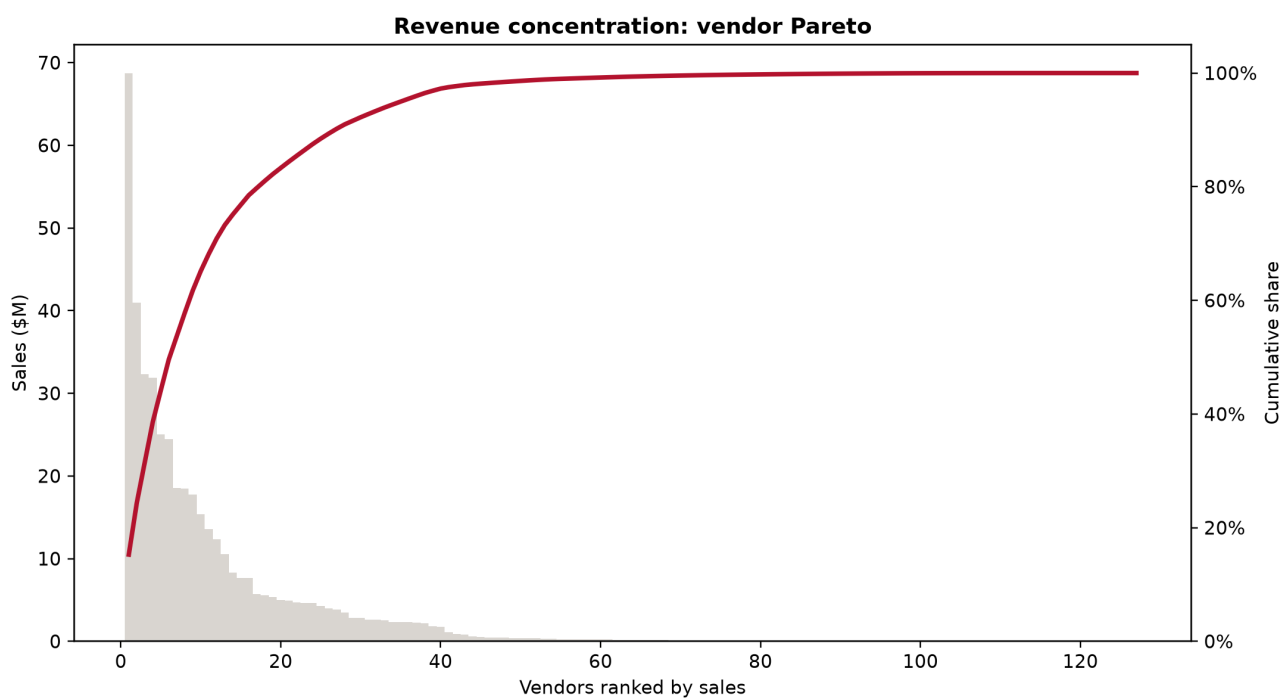
BUSINESS IMPLICATION

Pair sales rank with margin and turnover before promotional investment; a sales leader may still be economically weak.

LIMITATION

Brand descriptions may recur across vendors and are aggregated by Brand plus Description.

Pareto: concentration is actionable



Cumulative curve is calculated from vendors sorted by recorded sales.

VERIFIED RESULT

The top 10 vendors produce 65.2% of sales; the top 20 produce 83.3%.

INTERPRETATION

A relatively small supplier cohort carries a large share of revenue exposure.

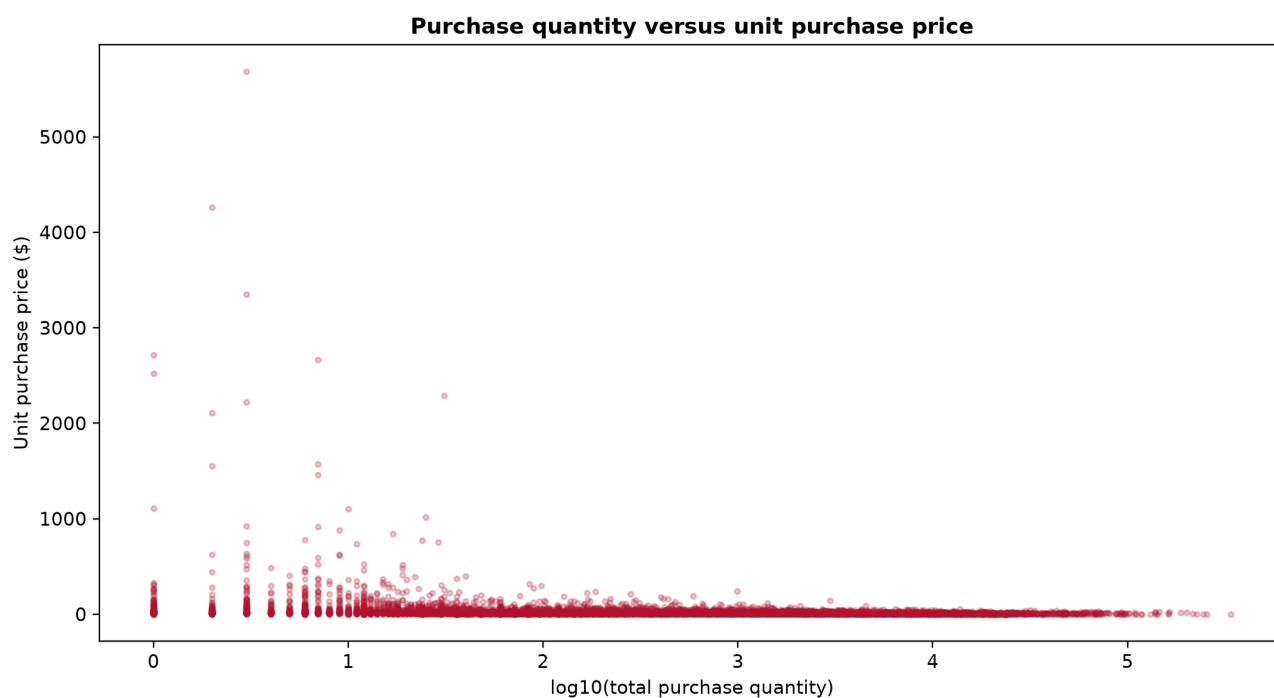
BUSINESS IMPLICATION

Prioritize forecast reviews, supply risk checks and commercial negotiations for this cohort.

LIMITATION

A Pareto curve does not reveal substitutability, contract terms or supplier lead times.

Bulk purchasing signal



Scatter uses log-scaled purchase quantity for readability; each point is a summary record.

VERIFIED RESULT

Spearman $\rho = -0.275$ ($p = 2.39e-184$) between purchase quantity and unit purchase price.

INTERPRETATION

The rank correlation measures monotonic association without assuming a linear price curve.

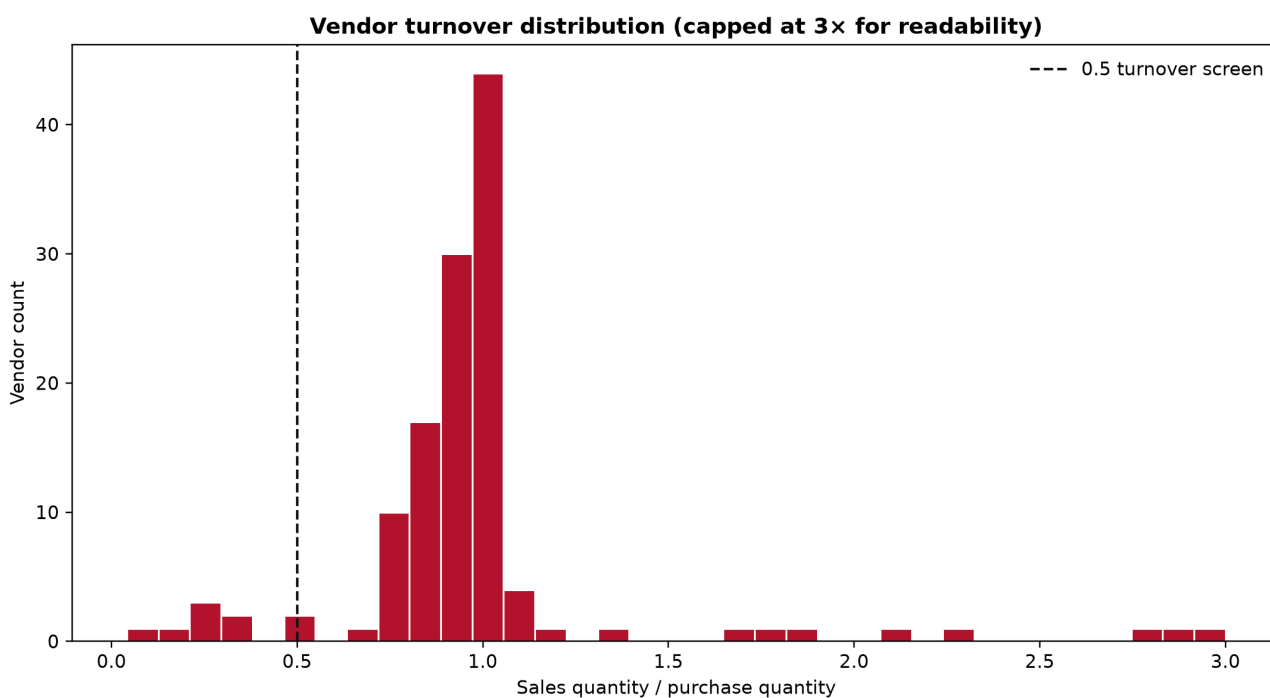
BUSINESS IMPLICATION

Treat observed volume-price patterns as a negotiation hypothesis; test matched products and contract terms before changing order sizes.

LIMITATION

Mixed products, package sizes and vendor effects confound a pooled correlation; it is not causal evidence of a bulk discount.

Inventory movement screen



Turnover equals total sales quantity divided by total purchase quantity; chart is capped at 3× only for visual readability.

VERIFIED RESULT

11 vendors fall below a 0.5 turnover screen, representing \$50,110 of recorded purchase dollars; median vendor turnover is 0.97×.

INTERPRETATION

Low turnover may indicate slow movement, overbuying, timing mismatch or incomplete sales matching.

BUSINESS IMPLICATION

Review the largest low-turnover exposures first and pause replenishment pending item-level validation.

LIMITATION

This is not days-of-supply: no inventory dates, on-hand balance reconciliation or demand seasonality is applied.

Statistical comparison

VERIFIED RESULT

High-sales records average 28.9% margin versus -60.4% for lower-sales records;
Welch t = 10.24, p = 2.13e-24.

INTERPRETATION

Groups split at median record-level sales of \$5,580; Welch's test compares mean margins while allowing unequal variances.

BUSINESS IMPLICATION

The result supports prioritizing a margin review by sales tier, not a blanket assumption that scale guarantees economics.

LIMITATION

Rows are not independent commercial experiments; skew, repeated vendors and product mix limit causal or population inference.

Decision matrix

VERIFIED RESULT

Use four views together: scale, margin, turnover and concentration.

INTERPRETATION

A vendor can be a sales leader yet weak on gross margin; a high-margin brand can be slow moving.

BUSINESS IMPLICATION

Create a weekly exception list: high sales/low margin, low turnover/high purchase dollars, and concentrated critical suppliers.

LIMITATION

Thresholds shown here are analytical screens, not approved business policy or service-level targets.

Recommendation: protect profitable scale

VERIFIED RESULT

Focus account governance on the top-10 cohort that drives 65.2% of sales.

INTERPRETATION

Concentration makes service disruption and pricing decisions consequential.

BUSINESS IMPLICATION

Assign executive ownership, forecast checks and margin guardrails to strategic vendors and brands.

LIMITATION

The data lacks contract, fill-rate and supplier-risk fields needed to operationalize a complete vendor scorecard.

Recommendation: buy with evidence

VERIFIED RESULT

The observed quantity–price association is $\rho = -0.275$, not a causal estimate.

INTERPRETATION

Pooled data suggests a pattern worth testing but cannot isolate product, pack size or negotiated terms.

BUSINESS IMPLICATION

Run controlled, SKU-matched quote comparisons before committing to bulk-buy changes.

LIMITATION

A correlation cannot quantify an expected savings rate or support a specific order-size recommendation.

Recommendation: manage movement

VERIFIED RESULT

Start with 11 low-turnover vendor screens, then drill to items with the largest purchase-dollar exposure.

INTERPRETATION

Quantity turnover identifies mismatch between recorded purchasing and sales movement.

BUSINESS IMPLICATION

Freeze or reduce replenishment only after validating on-hand stock, seasonality and recent sell-through.

LIMITATION

No inventory-age or stock-on-hand reconciliation is available in the summary table.

Metric definitions

VERIFIED RESULT

Gross profit = sales dollars – purchase dollars; gross margin = gross profit / sales dollars.

INTERPRETATION

Turnover = sales quantity / purchase quantity; sales-to-purchase = sales dollars / purchase dollars.

BUSINESS IMPLICATION

These formulas make the model transparent and reproducible from the SQLite summary.

LIMITATION

They are decision-support indicators, not final accounting measures; ratios can become undefined when denominators are zero.

Limitations & future work

VERIFIED RESULT

Freight is vendor-level, sales may be zero-filled after joins, and the source has no dated inventory-aging or service-level lens.

INTERPRETATION

These constraints limit product-level profitability and inventory conclusions.

BUSINESS IMPLICATION

Next: allocate landed cost, add inventory snapshots/age, returns, lead time, promotion flags and customer demand data.

LIMITATION

Future enrichments require source-system validation and a documented allocation methodology.

Technical stack & structure

VERIFIED RESULT

SQLite + SQL build the consolidated table; Python/pandas calculate derived metrics; SciPy performs statistical tests; Matplotlib produces charts; ReportLab renders this PDF.

INTERPRETATION

The project is deliberately local and inspectable: raw CSV → SQLite tables → vendor_sales_summary → notebook/report.

BUSINESS IMPLICATION

The workflow can run on a laptop and can be version-controlled with the analytical code.

LIMITATION

The source database is large and local; refresh time and source-file availability affect reproducibility.

Reproducibility appendix

VERIFIED RESULT

Run `ingestion_db.py`, then `get_vendor_summary.py`, then `build_case_study.py`; run `python -m unittest test_notebook_final_merge.py test_report_metrics.py` for regression checks.

INTERPRETATION

The report reads `inventory.db` / `vendor_sales_summary` and regenerates each added chart into `report_assets/`.

BUSINESS IMPLICATION

This report was built from the local SQLite snapshot present at render time; all displayed calculated figures are dynamically inserted.

LIMITATION

The notebooks are exploratory artifacts; their saved output may not contain all later analytical sections described in the README.